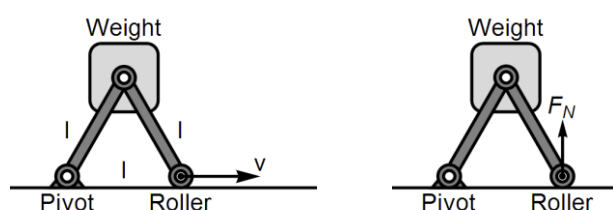


Chap. 18 Rotation in Two Dimensions

1. The Choo-choog Machine (IV): Torque and Angular Momentum

This will be the last time that we look at the choo-choog machine. A weight of mass M is held up by the elbow like pivoted rods of the lengths l . We have already known that to keep the roller moving with a constant speed v , there must be a horizontal force on the roller. However there must also be a vertical force F_N to support the roller on the floor. So we are going to find it.

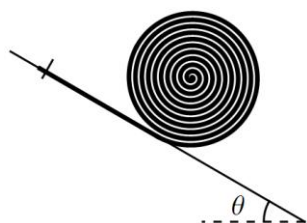


When the distance between the roller and the pivot is l ,

- What is the angular momentum L of the weight with respect to the pivot?
- What is the changing rate of the angular momentum dL/dt ?
- Consider the torque τ with respect to the pivot due to the forces exerted on the roller, and determine the force F_N .

2. Flexible Tape Rolling Down

A flexible tape of the length l is tightly wound. It is then allowed to unwind as it rolls down a steep incline that makes an angle θ with the horizon. The upper end of the tape is tacked down. Find the time it takes for the tape to unwind completely.



3. Hidden Angular Momentum in the Magnetic Field

Momentum $\mathbf{p}$ and angular momentum $\mathbf{L}$ are closely related by $\mathbf{L} = \mathbf{r} \times \mathbf{p}$. In the homework problem 3 of Chap. 10, we have discovered the potential momentum (i.e. the electromagnetic momentum) hidden in the magnetic field. Now it is time to generalize the potential momentum to the *potential angular momentum* (i.e. the electromagnetic angular momentum),

$$\mathbf{L}_{EM} = q\mathbf{r} \times \mathbf{A}.$$

Remember the vector potential $\mathbf{A}$ can be written as (by symmetric gauge)

$$\mathbf{A} = \frac{1}{2}\mathbf{B} \times \mathbf{r},$$

for uniform magnetic field $\mathbf{B}$.

(a) Consider a charge q cycling in a uniform magnetic field B on a circular orbit of radius r (magnetic field perpendicular to the orbital plane). What is the strength of vector potential A on the orbit? How much potential angular momentum will the charge processes on the orbital?

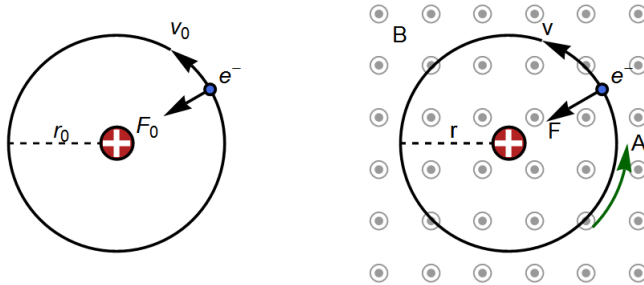
To give an example of the physical consequence of the potential angular momentum, let us look at the *Landau diamagnetic effect*. Diamagnetism is a very weak form of magnetism that is only exhibited in the presence of an external magnetic field. It is the result of changes in the orbital motion of electrons due to the external magnetic field. The induced magnetic moment is very small and in a direction opposite to that of the applied field. Diamagnetism is found in all materials; however, because it is so weak it can only be observed in materials that do not exhibit other forms of magnetism.

As a simple model, we consider an representative atom in the material. There is one electron orbiting around the nuclear on circular orbit of radius r_0 . The electron charge is $-e$ and the nuclear charge is e (like a hydrogen atom). The electron mass is m .

(b) What is the orbital velocity v_0 of the electron? What about the cycling period?

The magnetic moment of a current circuit is defined as $M = IS$ with its direction given by right-hand rule, where I is the current running in the orbit and S is the area enclosed by the orbit.

(c) Then what is the magnetic moment due to the electron's orbital motion around the nuclear?



We adiabatically apply the magnetic field B perpendicular to the electron orbital plane. Both the radius of the orbit and the velocity of the electron may change in the presence of the magnetic field. But the shape of the orbit still remains circular.

(d) What is the relation between the new orbital radius r and velocity v so as to keep the electron on a circular orbit? Hints: you need to take into account both the Coulomb force and Lorentz force.

(e) What is the (kinetic) angular momentum of the electron (with respect to the nuclear) before and after applying the magnetic field (in terms of r_0 , v_0 , r and v)? How much potential angular momentum does the electron have in the magnetic field?

(f) The conservation of angular momentum states that the total angular momentum keeps the same before and after applying the magnetic field. Use this law to relate r and v to r_0 and v_0 .

In the weak field limit $B \rightarrow 0$, the orbital radius r and velocity v does not change much from r_0 and v_0 , thus they can be expanded around r_0 and v_0 by $r = r_0 + \delta r$ and $v = v_0 + \delta v$. Treat δr and δv as very small quantities of the same order as B .

(g) Determine δr and δv by first order perturbation. Calculate the corresponding change in the magnetic momentum. Hint: Taylor expand both sides of your equations to the first

order of δr , δv and B , and solve the linear equation.

The *magnetic susceptibility* for the single atom is defined as

$$\chi = \mu_0 \frac{\partial M}{\partial B},$$

where $\mu_0 = 4\pi \times 10^{-7} \text{ N} \cdot \text{A}^{-2}$ is the magnetic constant (the vacuum permeability).

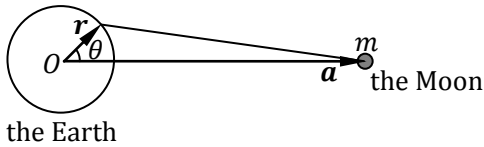
(h) According to this definition, find the magnetic susceptibility of the atom.

The negative susceptibility means the applied magnetic field will induce a magnetic moment in the opposite direction, which in turn weakens the magnetic field in the material, hence the diamagnetism.

(i) Make a rough estimate of the diamagnetic susceptibility of N_2 gas in the air (the overall susceptibility per unit volume), given the molecule size of N_2 $r_0 = 155 \text{ pm}$ and its mass density in the atmosphere $\rho = 0.91 \text{ g} \cdot \text{m}^{-3}$.

4. Tidal Force and Tidal Phenomena

Tidal force is the cause of tides. It mainly comes from the Moon and the Sun. Let us first consider the influence of the Moon. As shown in the following figure, the Moon m is away from the Earth center by the displacement a .



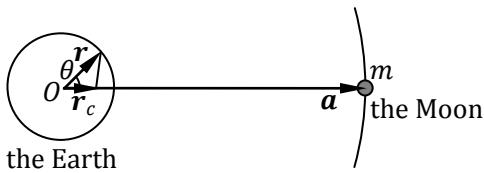
(a) What is the gravitational potential energy $V_m(\mathbf{r})$ at the surface of the Earth $\mathbf{r}$, corresponding to the gravitational field of the Moon.

The Earth-Moon distance a is about sixty times of the radius r of the Earth. To approximate the Moon's gravitational potential on the Earth's surface, we can Taylor expand $V_m(\mathbf{r})$ with respect to (r/a) up to the second order

$$V_m(\mathbf{r}) = V_{m0} + V_{m1} \frac{r}{a} + V_{m2} \left(\frac{r}{a} \right)^2 + \dots$$

(b) Find V_{m0} , V_{m1} and V_{m2} as functions of the angle θ .

We should not forget that the Earth-Moon system is rotating about its center of mass $\mathbf{r}_c$. Suppose the Moon's orbital is circular, and its angular velocity is ω .



(c) Where is the center of mass of the Earth-Moon system? (Find the length of r_c in terms of G , m , ω and a)

Both the Earth and the Moon are rotating with respect to their center of mass (COM). Therefore the Earth itself is a non-inertial frame (even if we forget about the spinning of the Earth), in which all objects will experience the inertial force. If the object is not moving with high speed, the inertial force it experienced is basically the centrifugal force.

(d) What is the potential $V_\omega(\mathbf{r})$ corresponding to the centrifugal force due to rotation of the Earth with respect to the COM.

Now let us put $V_m(\mathbf{r})$ and $V_\omega(\mathbf{r})$ together with the potential of the Earth itself $V_M(\mathbf{r}) = -GMr^{-1}$ (where M is the mass of the Earth), thus the total potential on the surface of the Earth is

$$V(\mathbf{r}) = V_M(\mathbf{r}) + V_m(\mathbf{r}) + V_\omega(\mathbf{r}).$$

Make the Taylor expansion again,

$$V(\mathbf{r}) = -\frac{GM}{r} + V_0 + V_1 \frac{r}{a} + V_2 \left(\frac{r}{a}\right)^2 + \dots$$

The term V_0 is a constant which contributes nothing to the strength of the gravity field.

(e) Show that V_1 vanishes due to the cancelation between $V_m(\mathbf{r})$ and $V_\omega(\mathbf{r})$.

(f) Find out V_2 , which should be a function of θ only.

Now the total potential can be simply written as

$$V(\mathbf{r}) = V_M(\mathbf{r}) + V_{\text{tidal}}(\mathbf{r}),$$

where $V_M(\mathbf{r})$ is the gravitational potential of the Earth, and $V_{\text{tidal}}(\mathbf{r}) = V_2(r/a)^2$ is the potential that give rise to the tidal force. It is clear that $V_{\text{tidal}}(\mathbf{r})$ origins from the gravitay of the Moon.

The gravitational field strength on the surface of the Earth can be obtained from the potential by $\mathbf{g} = -\nabla V(\mathbf{r})$,

$$\mathbf{g} = g_r \mathbf{e}_r + g_\theta \mathbf{e}_\theta.$$

(g) Find out g_r and g_θ .

g_r contains both the free-falling acceleration and the radial tidal acceleration, while g_θ is the transverse tidal acceleration. They are responsible for the tidal phenomena. Unlike the gravity force, the tidal force is inversely proportional to the cube (not square) of the Earth-Moon distance.

(h) Given that $M = 5.98 \times 10^{24} \text{kg}$, $m = 7.36 \times 10^{22} \text{kg}$ and $a/r = 60.3$, how many orders of magnitude is the tidal force smaller than the gravity force on the Earth's surface.

Suppose the Earth is a sphere of the radius R . Let h be the sea level (i.e. $r = R + h$ is the distance of the ocean surface to the center of the Earth). The surface of the ocean is an equal-potential surface given by $mg_r h = \text{const}$.

(i) Plot the sea level h as a function of θ . How many times of flood tides are there in a day?

Chap. 19 Center of Mass; Moment of Inertia

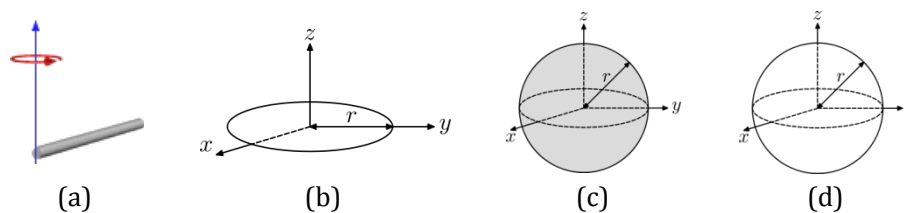
1. Finding the Moment of Inertia

(a) Rod of length L and mass m . Find the moment of inertia about the axis of rotation at the end of the rod.

(b) Thin circular disk of radius r and mass m . Find the moments of inertia I_x , I_y and I_z about the three axes respectively.

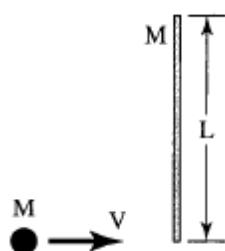
(c) Solid sphere of radius r and mass m . Find the moment of inertia about the axis through the center.

(d) Hollow sphere of radius r and mass m . Find the moment of inertia about the axis through the center.



2. Flying Putty Sticks to a Rod (I)

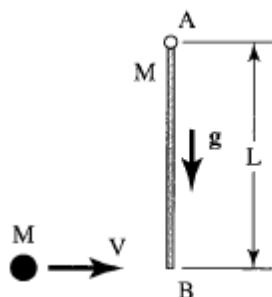
A thin rod of mass M and length L rests on a horizontal frictionless surface. A small piece of putty, also of mass M , and with velocity v directed perpendicularly to the rod, strikes one end and sticks, making an inelastic collision of very short duration.



- What is the velocity of the center of mass of the putty-rod system before and after the collision.
- What is the angular momentum of the putty-rod system about its center of mass just before the collision.
- What is the angular velocity (about the center of mass) just after the collision?
- How much kinetic energy is lost in the collision?

3. Flying Putty Sticks to a Rod (II)

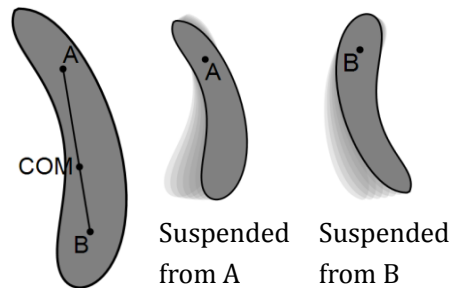
A thin uniform rod AB of mass M and length L is free to rotate in a vertical plane about a horizontal axle at the end A. A piece of putty also of mass M is thrown with velocity v horizontally at the lower end B while the bar is at rest. The putty sticks to the bar. What is the minimum velocity of the putty before impact that will make the bar rotate *all the way* around A?



4. Physical Pendulum

The physical pendulum in the following figure has two possible pivot points A and B. Point A has a fixed position, and point B is adjustable along the line through point A and the center of

mass (COM) of the pendulum. The period of the pendulum when suspended from A is found to be T . The pendulum is then reversed and suspended from B, which is moved until the pendulum again has the period T . If the distance between A and B is found to be l for equal periods, what is the freefall acceleration g at the pendulum's location?



Note: this method can be used to measure g without knowledge of the rotational inertia of the pendulum or any of its dimensions except l .

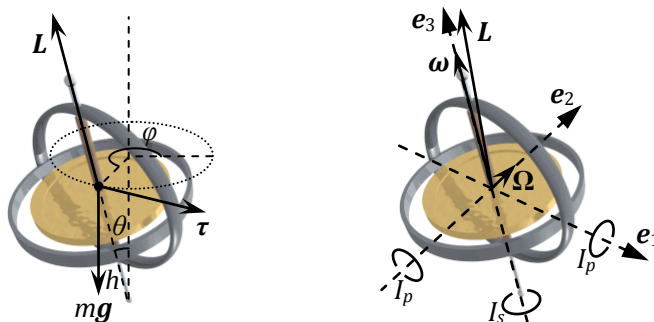
Chap. 20 Rotation in Space

1. Precession and Nutation of Gyroscope

A gyroscope exhibits a number of behaviors including precession and nutation. The fundamental equation describing the behavior of the gyroscope is:

$$\boldsymbol{\tau} = \frac{d\mathbf{L}}{dt},$$

where the vectors $\boldsymbol{\tau}$ and $\mathbf{L}$ are the torque on the gyroscope and its angular momentum respectively. The torque is supplied by a couple of forces: gravity acting downwards on the device's centre of mass, and an equal force acting upwards to support one end of the device.



(a) Consider the gyroscope of mass m with its spinning axis inclined from the vertical direction by the angle θ . If its center of mass is h distant away from the supporting end, what is the torque exerted on the gyroscope (with respect to the supporting end)?

(b) If the magnitude of the angular momentum of the spinning gyroscope is L , find the angular velocity of the precession Ω_p .

Due to the precession motion, the angular momentum does not exactly falls along the direction of the spinning axis. To show this, let us break the angular velocity vector into components $\boldsymbol{\omega}$ and $\boldsymbol{\Omega}$ perpendicular and parallel to the spinning wheel of the gyroscope. The physical meaning of $\boldsymbol{\omega}$ is the angular velocity of the wheel, while $\boldsymbol{\Omega}$ stands for the angular velocity of the axis (as the axis rotating about the supporting end). With this

decomposition, the precession and the nutation motion is separated from the general motion of the gyroscope.

$\mathbf{e}_1$, $\mathbf{e}_2$ and $\mathbf{e}_3$ are the unit vectors along the three principle axes (one of which is the spinning axis and the other two lies in the plane of the wheel). The moments of inertia about the principle axes are different: I_s is the moment of inertia about the spinning axis, and I_p is moment of inertia about either of the other two perpendicular principal axes. The two I_p 's should be almost the same, due to the approximate symmetry of the gyroscope.

(c) So what is the angular momentum $\mathbf{L}$ in terms of the angular velocities $\boldsymbol{\omega}$ and $\boldsymbol{\Omega}$? Is it parallel to the spinning axis?

For a realistic gyroscope, the precession or nutation motion of the axis is much slower compared to the rapid spinning of the wheel about the axis. This is to say that ω is usually orders of magnitude greater than Ω , and is almost a constant independent of the motion of the axis.

(d) By carry out the time derivative $d\mathbf{L}/dt$, prove that the equation of motion becomes,

$$\boldsymbol{\tau} = I_s \boldsymbol{\Omega} \times \boldsymbol{\omega} + I_p \frac{d\boldsymbol{\Omega}}{dt}.$$

Hints: the vector $\boldsymbol{\omega}$ only changes in its direction, and its magnitude is kept constant.

If we denote the component of angular momentum along the spinning axis by $L_0 = I_s \omega$, then the above equation can be casted into

$$\boldsymbol{\tau} = \boldsymbol{\Omega} \times \mathbf{L}_0 + I_p \frac{d\boldsymbol{\Omega}}{dt}.$$

We have learned from the Lecture that the first term on the right-hand-side describes the precession. The extra second term on the right-hand-side will give rise to the nutation.

However we are not very interested in the dynamics of $\boldsymbol{\Omega}$ itself. What we concern more is the dynamics of the axis of the gyroscope. Let us define the unit vector of the axis by $\mathbf{n} \equiv \boldsymbol{\omega}/\omega$. The axis rotates about the supporting end with the angular velocity $\boldsymbol{\Omega}$, so that

$$\frac{d\mathbf{n}}{dt} = \boldsymbol{\Omega} \times \mathbf{n}.$$

(e) Show that $\boldsymbol{\Omega}$ can be "solved" from the above relation,

$$\boldsymbol{\Omega} = \mathbf{n} \times \frac{d\mathbf{n}}{dt}.$$

(f) Substituting into the equation of motion for $\boldsymbol{\Omega}$, show that $\mathbf{n}$ satisfies the following equation of motion,

$$I_p \frac{d^2 \mathbf{n}}{dt^2} = \boldsymbol{\tau} \times \mathbf{n} - \frac{d\mathbf{n}}{dt} \times L_0.$$

However, we are not going to solve this equation directly, because it has no analytic solution in general, and we do not wish to see our homework becoming an absolute miracle. Let us make some physical interpretations. Recall the equation of motion for a charge in the electromagnetic field, which reads $m\mathbf{a} = \mathbf{F} = q\mathbf{E} + q\mathbf{v} \times \mathbf{B}$, or written as

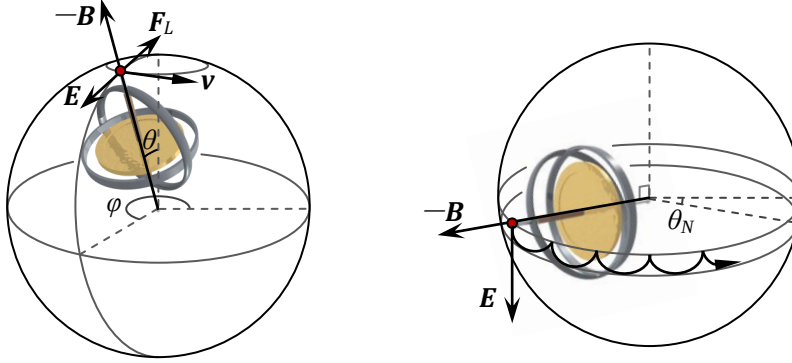
$$\frac{m}{q} \frac{d^2 \mathbf{x}}{dt^2} = \mathbf{E} + \frac{d\mathbf{x}}{dt} \times \mathbf{B}.$$

Compare with the equation we have just obtained for $\mathbf{n}$, we can make the following analog,

Axis Dynamics	Charge Dynamics
Position of the free end: $\mathbf{n}$	Position of the charge: $\mathbf{x}$

Moment of inertia: I_p	Inertia mass per charge: m/q
Force projected to the sphere: $\boldsymbol{\tau} \times \mathbf{n}$	Electric field: $\mathbf{E}$
Angular momentum of spinning: $-\mathbf{L}_0$	Magnetic field: $\mathbf{B}$

The dynamics of the free end of the gyroscope is just like a charge moving in a electric field $\mathbf{E} = \boldsymbol{\tau} \times \mathbf{n}$ and a magnetic field $\mathbf{B} = -\mathbf{L}_0$. The only difference is that the motion of is constrain on a unit sphere (known as the *Bloch sphere*), which is a 2-dimensional *curved space*.



(g) Calculate the magnitude and describe the direction of $\mathbf{E}$ and $\mathbf{B}$ for all range of θ and ϕ . If the electric force is balanced by the magnetic force, the gyroscope will not fall but precess with steady velocity $d\mathbf{n}/dt$. For a given latitude θ , determine the precession speed $|d\mathbf{n}/dt|$. Is it the same as the precession speed Ω_p you have obtained in the problem (b)?

Now let us understand the physical meaning of this electromagnetic field. The gravity always wants to lay down the gyroscope. Projecting the gravitational force onto the Bloch sphere leads to the electric field. So the electric field reflects the tendency of falling of the gyroscope. Then what about the magnetic field? As the wheel of the gyroscope starts to spin, the magnetic field will be set up over the Bloch sphere as if there were a magnetic monopole lying at the center of the sphere. The strength of the magnetic field is proportional to the angular velocity ω of the wheel. Therefore the magnetic field is an effect brought by the spinning of the wheel. Without the spinning, the gyroscope will surely fall as any of other objects in the gravitational field. But with the spinning, the falling of the gyroscope leads to the acceleration in the precession direction. This exactly mimic the effect of Lorentz force that the velocity in one direction generates the force in the perpendicular direction. It is just because of this magnetic field effect that the gyroscope could start precession after release.

Now it is easier to understand nutation. If the spinning gyroscope is hold still horizontally ($\theta = \pi/2$) and then release, it is just like releasing a rest charge in the electromagnetic field. Since the electric and magnetic field are perpendicular to each other, the motion of the charge will trace out a cycloid (the curve that is generated by a point on the circumference of a circle as it rolls along a straight line). To constrain the motion on the Bloch sphere, the cycloid should be bent around the equator, thus giving the track of nutation motion.

(h) According to the above analysis, calculate the amplitude (the nutation angle θ_N) and the period of nutation, in the large ω limit.

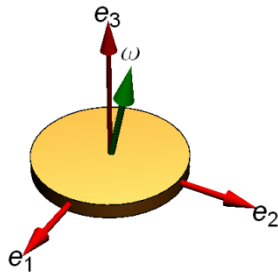
2. Torque-Free Precession of a Disk

Torque-free precession occurs when the axis of rotation differs slightly from an axis about which the object can rotate stably: a maximum or minimum principal axis. For example,

when a plate is thrown, the plate may have some rotation around an axis that is not its axis of symmetry. This occurs because the angular momentum $\mathbf{L}$ (the longer arrow) is constant in absence of torques. Therefore it will have to be constant in the external reference frame, but the moment of inertia tensor $\mathbf{I}$ is non-constant in this frame because of the lack of symmetry. Therefore the spin angular velocity vector $\boldsymbol{\omega}_s$ (the shorter arrow) about the spin axis will have to evolve in time so that the matrix product $\mathbf{L} = \mathbf{I}\boldsymbol{\omega}_s$ remains constant.



Among the three principle axes of a disk, one ($\mathbf{e}_3$) is perpendicular to the disk, and the other two ($\mathbf{e}_1$ and $\mathbf{e}_2$) lies in the plane of the disk. Let I_s be the moment of inertia about the $\mathbf{e}_3$ axis, and I_p be moment of inertia about either of the other two principal axes. They should be the same, due to the symmetry of the disk.



At any instance, the three principle axes span a rotating frame of reference (in which the disk looks static for the moment) known as the frame of principle axes.

(a) If the angular velocity in the frame of principle axes is $\boldsymbol{\omega} = \omega_1\mathbf{e}_1 + \omega_2\mathbf{e}_2 + \omega_3\mathbf{e}_3$, what is the angular momentum $\mathbf{L} = L_1\mathbf{e}_1 + L_2\mathbf{e}_2 + L_3\mathbf{e}_3$? (Give the expression for L_1 , L_2 and L_3)

Because the frame of principle axes is rotating with the angular velocity $\boldsymbol{\omega}$, its base vector will change with respect to time according to $\dot{\mathbf{e}}_i = \boldsymbol{\omega} \times \mathbf{e}_i$ in the external inertial reference frame. Since there is no torque exerting on the disk, we should have

$$\frac{d\mathbf{L}}{dt} = \boldsymbol{\tau} = 0$$

in the external frame.

(b) Show that the above equation leads to Euler's equation in the frame of principle axes,

$$\frac{\tilde{d}\mathbf{L}}{\tilde{d}t} + \boldsymbol{\omega} \times \mathbf{L} = 0,$$

where the derivative $\tilde{d}/\tilde{d}t$ only applies to the components of the vector $\mathbf{L}$ in the frame of principle axes.

The torque-free precession will occur if the angular velocity $\boldsymbol{\omega}$ neither falls on the $\mathbf{e}_3$ axis nor lies in the plane of $\mathbf{e}_1$ and $\mathbf{e}_2$.

(c) Let α be the angle between $\boldsymbol{\omega}$ and $\mathbf{e}_3$, show that the angular velocity of torque-free precession is

$$\Omega = \omega(I_s/I_p - 1) \cos \alpha.$$